ENGINEERING DRAWING
EDR 101
SOLID OBJECT

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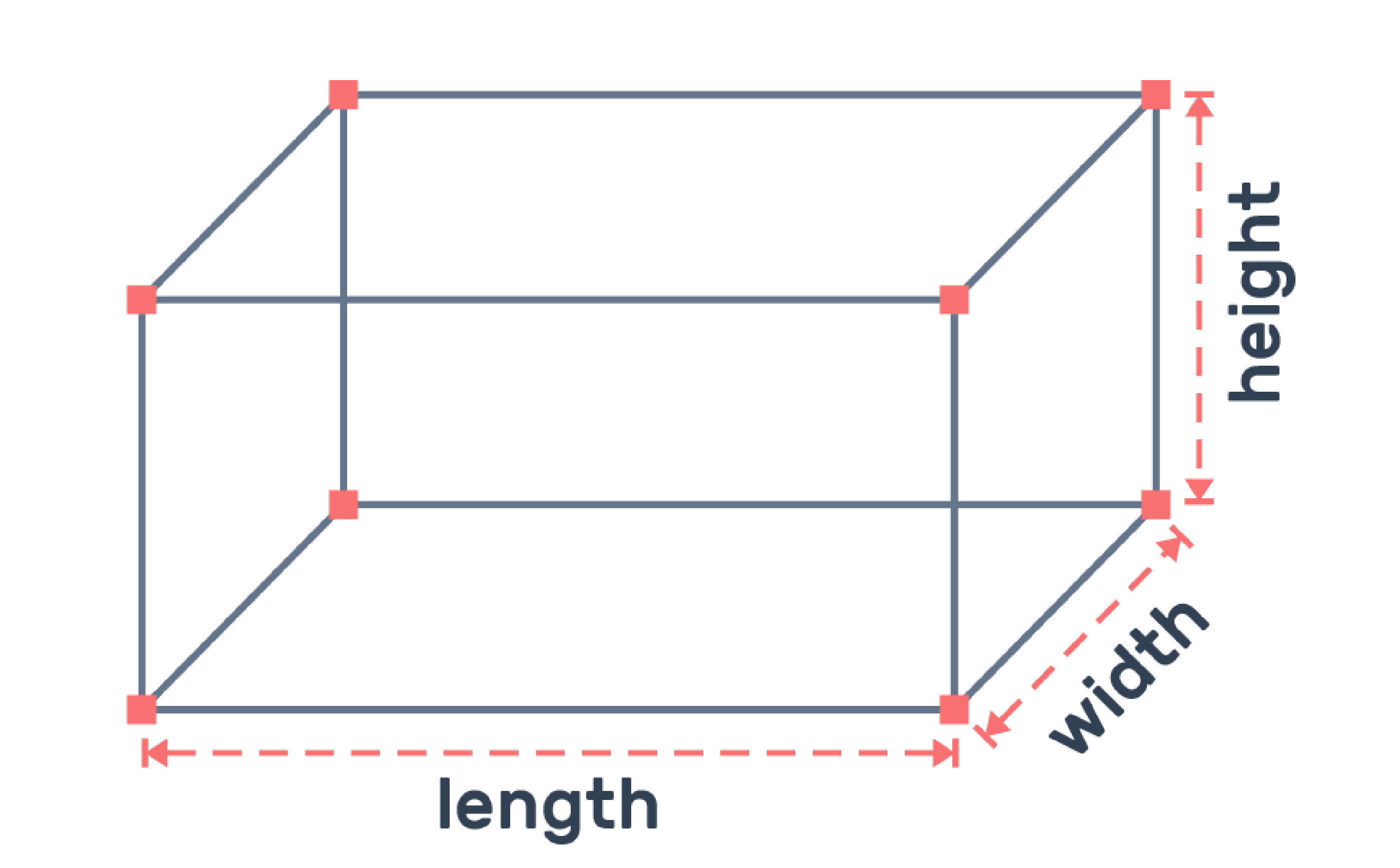
COURSE CONTENT

Lecture	Topics
01	Introduction to Engineering Drawing
02	Lettering, Numbering, Dimensioning, Drawing Scales
03	Plane Geometry: Pentagon, Hexagon, Octagon, Ellipse, Parabola, Hyperbola
04	Solid Object: Polyhedra, Prism, Pyramid, Cone, Cylinder, Sphere, Frustum, Truncated
05	Projection: Orthographic, Isometric, Oblique
06	Drawing isometric projections of solids; Drawing isometric views of objects
MID TERM – I	
07	Drawing orthographic projections of solids and different objects
08	Introduction to AutoCAD: Interface, Basic command, Layout, Creating different types of drawing, Dimension & Annotation, Printing & Publishing
09	Course wrap-up and evaluation
FINAL EXAMINATION	

WHAT IS SOLID?

An object having three dimensions, i.e., length, breadth, and height or thickness, is called a SOLID



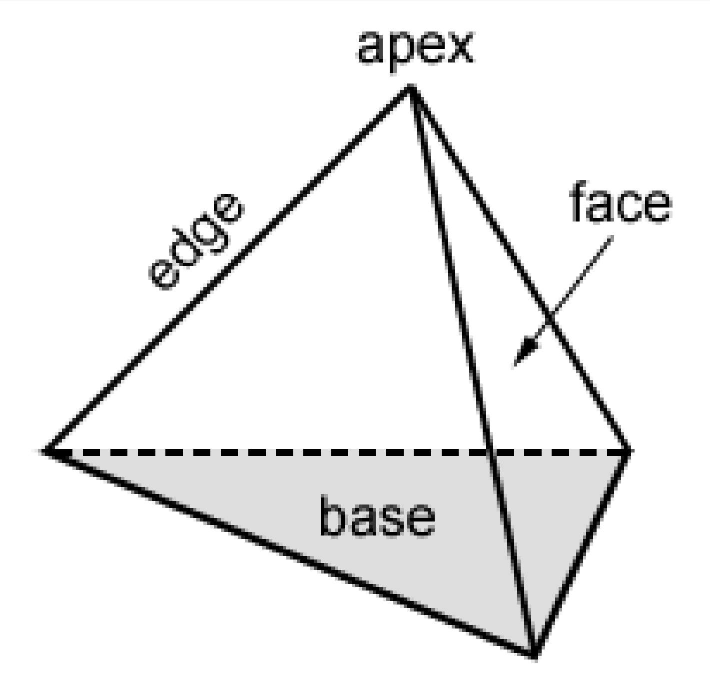
POLYHEDRA

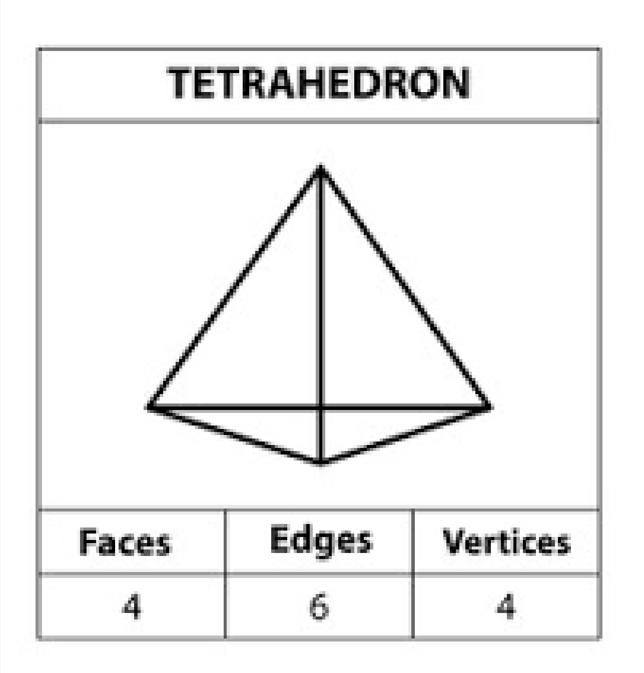
The solid which is bounded by plane surfaces or faces is called Polyhedron. The polyhedral are further sub-divided into three groups:

- Regular Polyhedra
- Prisms
- Pyramids

REGULAR POLYHEDRA

A polyhedron is regular if each of its plane surfaces is a **Regular Polygon**. The regular plane surfaces that form the surfaces of the polyhedra are called **Faces**. The lines at which two faces intersect are called **Edges**.





TYPES OF REGULAR POLYHEDRA

The 5 Regular Polyhedra -

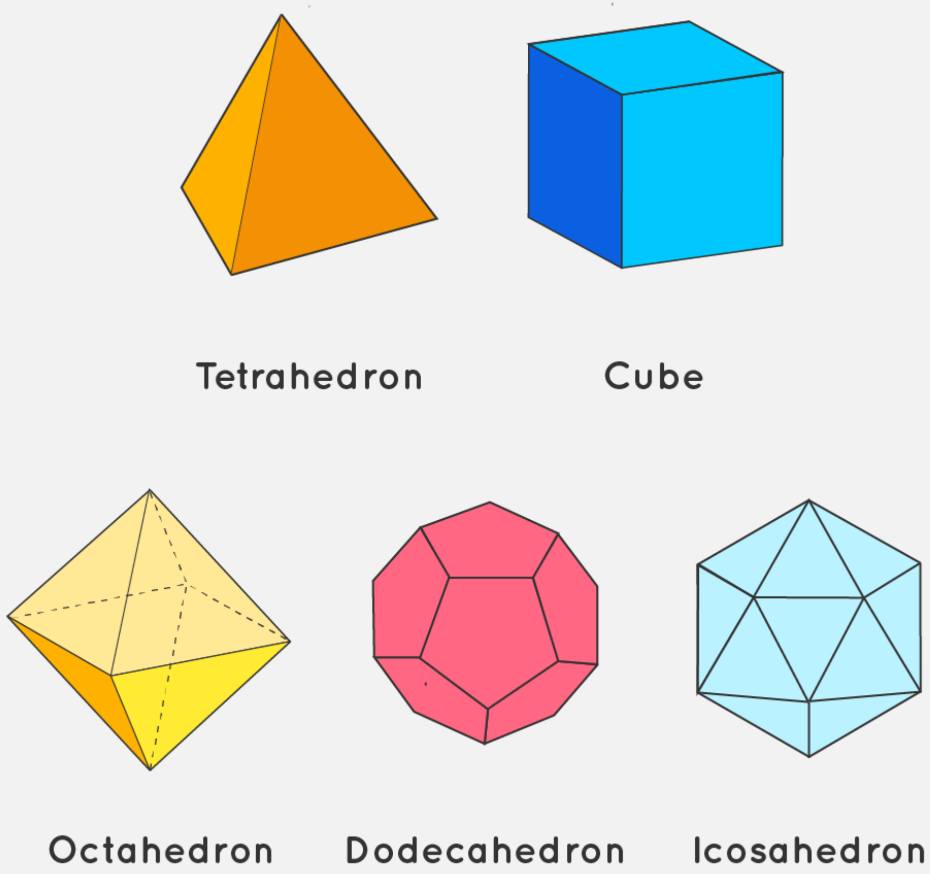
Tetrahedron: 4 faces (equilateral triangles)

Hexahedron (Cube): 6 faces (squares)

Octahedron: 8 faces (equilateral triangles)

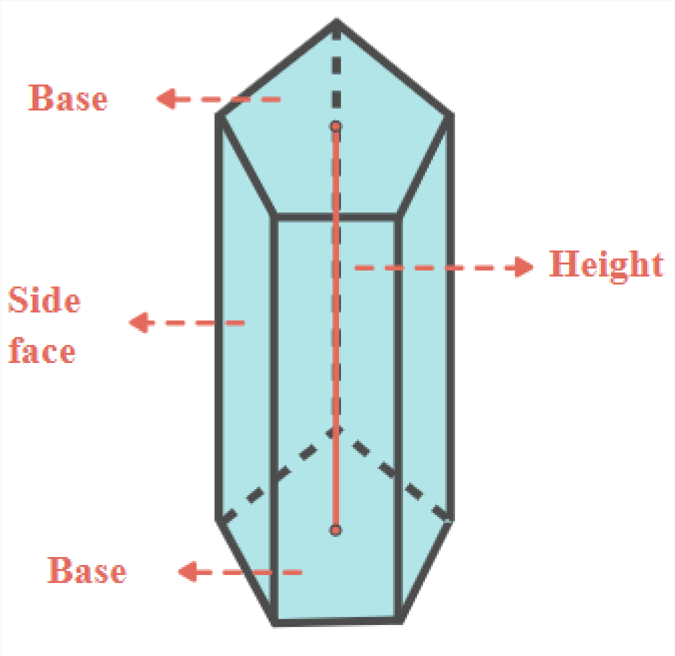
Dodecahedron: 12 faces (regular pentagons)

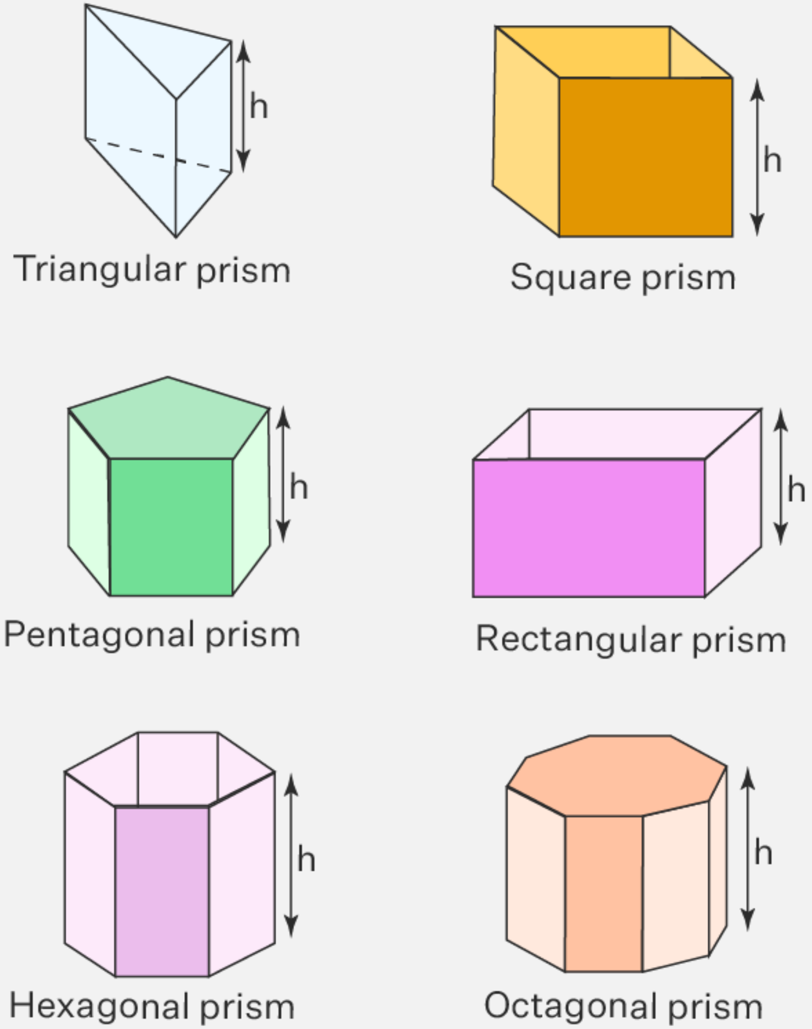
Icosahedron: 20 faces (equilateral triangles)



WHAT IS PRISM?

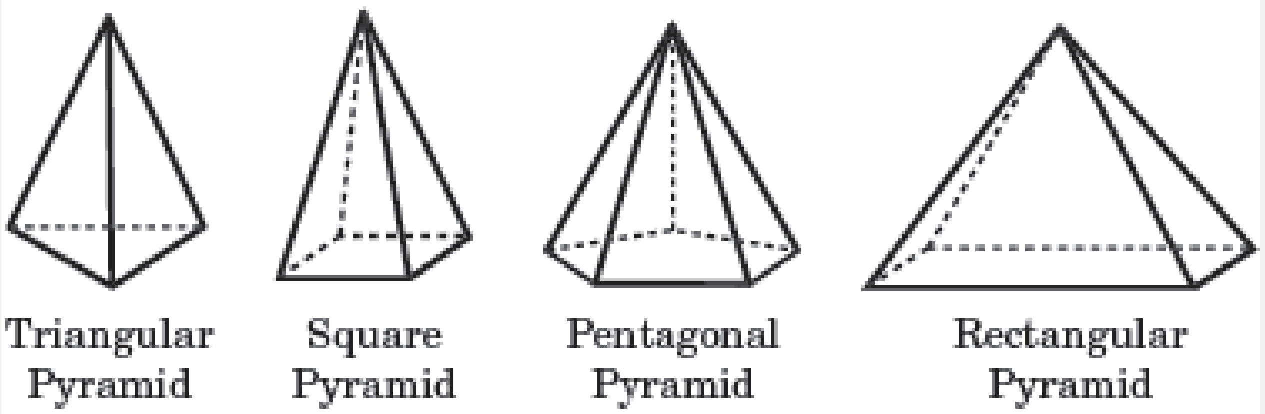
A solid figure whose bases or ends have the same size and shape and are parallel to one another, and each of whose sides is a parallelogram

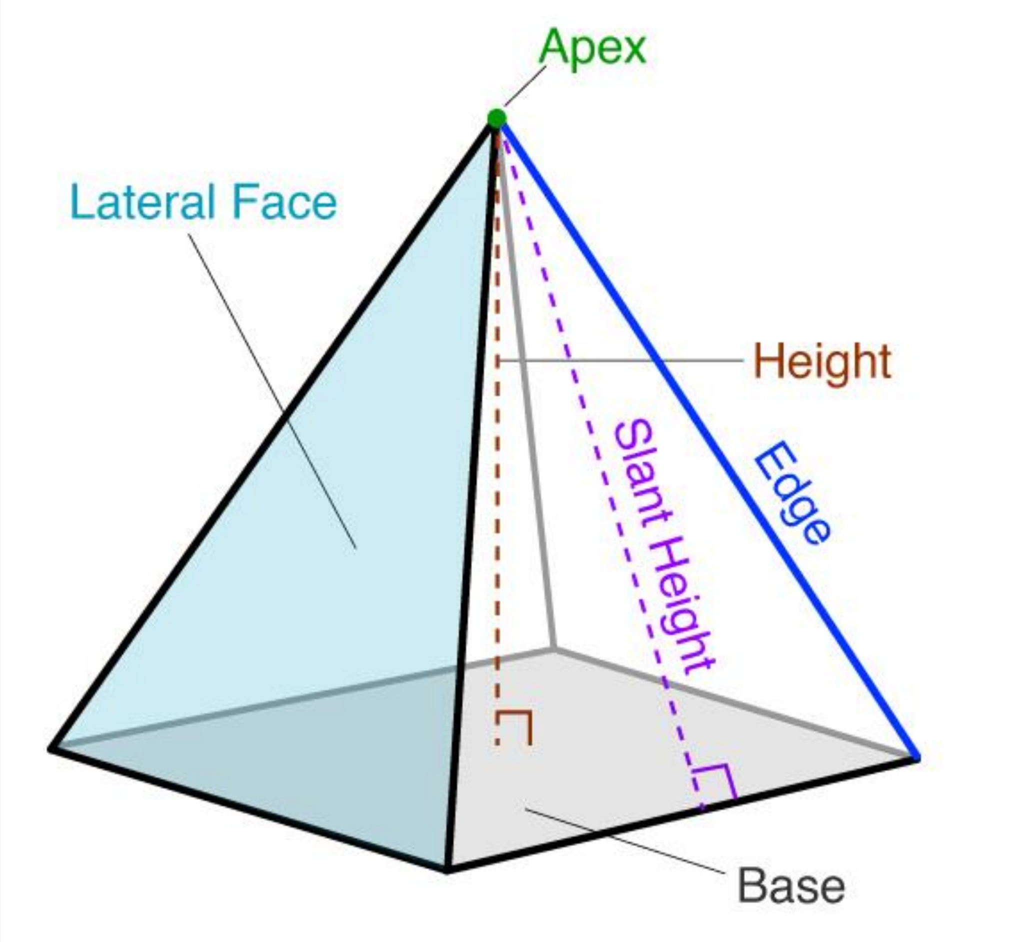




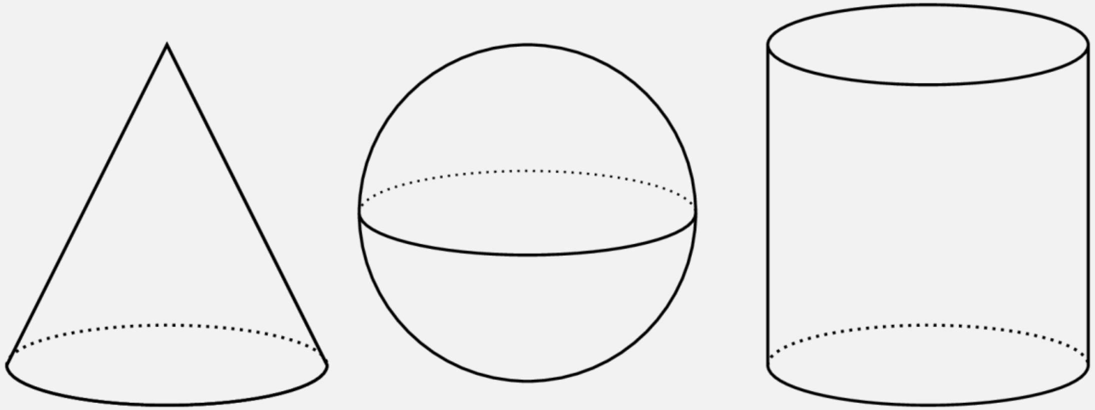
WHAT IS PYRAMID?

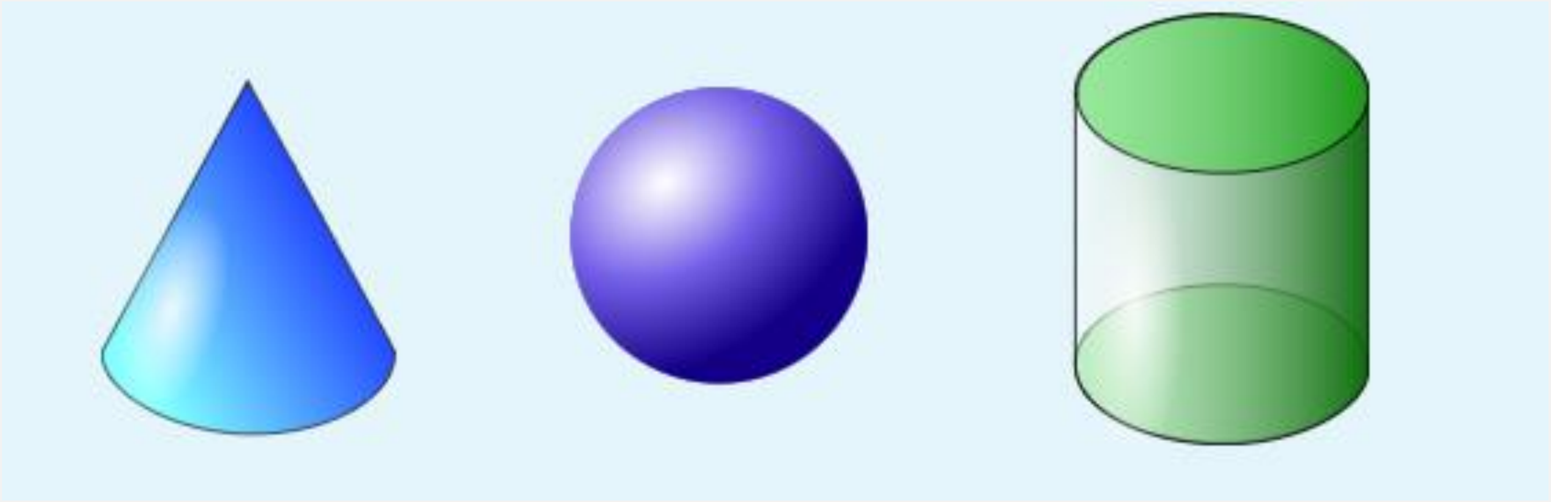
A solid figure with a polygonal base and triangular faces that meet at a common point called the apex





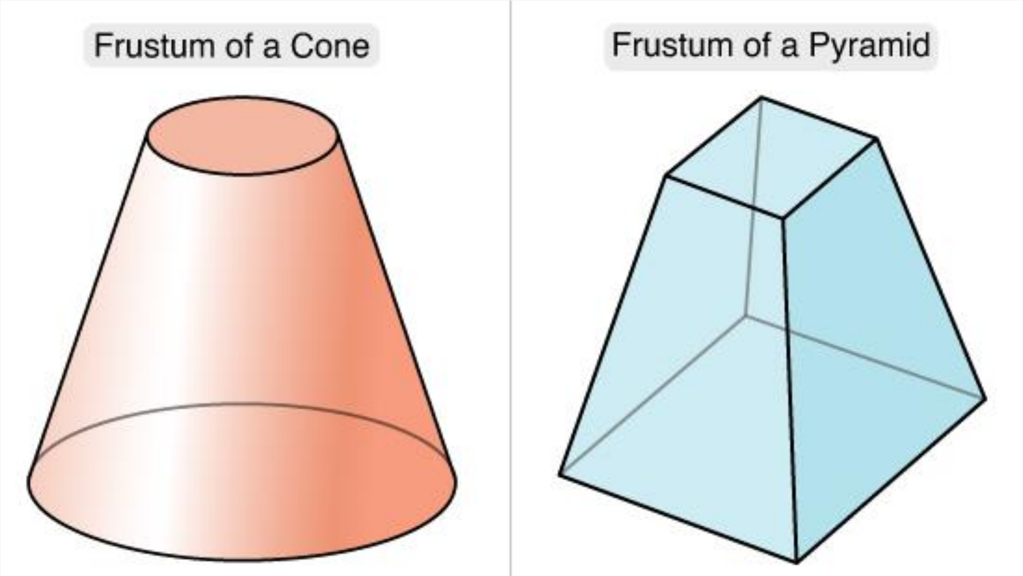
CONE, SPHERE, CYLINDER (SOLIDS OF REVOLUTION)

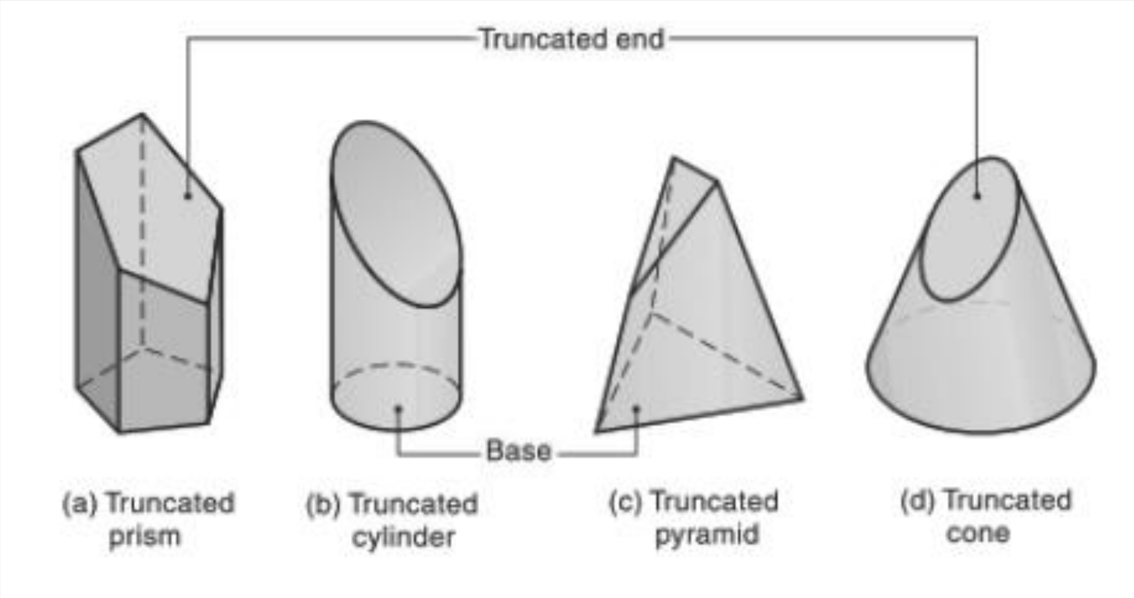




FRUSTUM AND TRUNCATED

When a pyramid or a cone is cut by a cutting plane parallel to its base, the remaining portion thus obtained after removing the top portion is called the **Frustum**





When a solid (polyhedron/cylinder/cone) is cut by a cutting plane inclined to its base (not parallel), the remaining portion thus obtained after removing the top portion is called the **Truncated Solid**

THANK YOU